Week 4 More on the convex geometry approach

Definition:
- A set $K \subseteq \mathbb{R}^d$ is a cone if
$$\forall \lambda \geq 0, \quad \lambda K \subseteq K \quad \forall x \in K, \lambda x \in K$$
- $K \subseteq \mathbb{R}^d$ is convex, if
$$\forall \alpha \in [0,1], x, y \in K, \alpha x + (1-\alpha) y \in K$$
- K is closed if K^c is open.

Defn: Let K be a close convex cone, define it statistical dimension:
$$\delta(K) := \mathbb{E}[\| P_K(G) \|_2^2]$$
$G \sim N(0, I_d)$, $P_K(G) := \operatorname{argmin}_{x \in K} \|G - x\|_2$.

Remarks:
- equivalent defn of $\delta(K)$ be given later in the proof sketch.
- we have the following (see an Exercise):
$$w^2(K \cap S^{d-1}) \leq \delta(K) \leq w^2(K \cap S^{d-1}) + 1$$
- If K is a linear subspace, $\delta(K) = \dim(K)$.

Thm (ALMT 14): Fix $\eta \in (0,1)$. Let C and K be two closed convex cones in $\mathbb{R}^d$. Let $Q \in \mathbb{R}^{d\times d}$ random orthogonal matrix. Then

$$\delta(C) + \delta(K) \le d - a_\eta \sqrt{d} \Rightarrow \mathbb{P}[C \cap QK \neq \{0\}] \le \eta \quad (1)$$

$$\cdots\cdots \ge d + a_\eta \sqrt{d} \Rightarrow \cdots\cdots \ge 1-\eta \quad (2)$$

where $a_\eta := \sqrt{8 \log 4/\eta}$

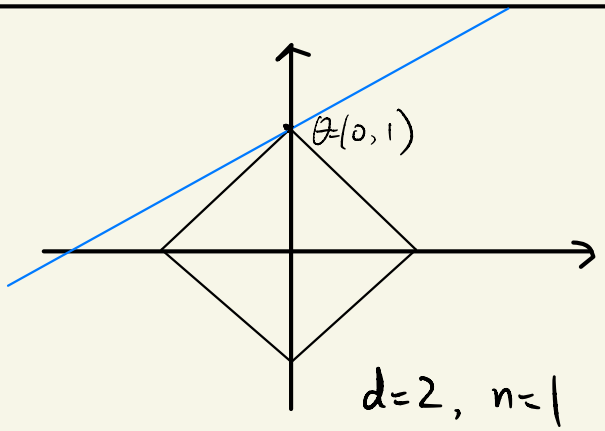
Remark: • (1) essentially covered by Gordon's theorem

• Statiscal dimension alone determines the intersection probability!

• phase transition with window size $\sim \sqrt{d}$.

Defn: $\theta \in \mathbb{R}^d$, define <u>descent cone</u>

$$D(\theta) := \{\delta \in \mathbb{R}^d : \exists \varepsilon > 0, \|\theta + \varepsilon\delta\|_1 \le \|\theta\|_1\}$$



$\mathbb{P}[\theta \text{ is exactly recovered}] = \frac{1}{2}$

Recall last time:

Given X, k, BP exactly recovers all k-sparse vectors

$$\Leftrightarrow \left(\bigcup_{\theta : k\text{-sparse}} D(\theta) \right) \cap N(X) = \{0\}$$

This time:

Given X, θ, BP exactly recovers θ

$$\Leftrightarrow \quad D(\theta) \cap N(X) = \{0\}$$

Proposition: Suppose $X \in \mathbb{R}^{d \times d}$, $N(X)$ is a (uniformly) random $(d-n)$-subspace. $\theta \in \mathbb{R}^d$ is k-sparse. Then $\exists$ $c, C > 0$, such that:

$$n \geq C\left(k \log d + \sqrt{d \log \frac{1}{\eta}}\right) \Rightarrow \mathbb{P}[\text{BP recovers } \theta] \geq 1 - \eta$$

$$n \leq c\left(k \log d - \sqrt{d \log \frac{1}{\eta}}\right) \Rightarrow \mathbb{P}[\text{BP recovers } \theta] \leq \eta$$

- We need $n \gtrsim k \log d$ measurements in order to exactly recover θ.

- ALMT 14 contains more examples of applications to demixing, low-rank matrix recovery.

Proof ideas :

- Intrinsic volume random variable

1st definition : · K is a polyhedral cone (intersection of finitely many half spaces)

$$\tilde{V}_j(K) := \mathbb{P}[P_K(G) \text{ is in the relative interior of a } j\text{-dimensional face of } K]$$

($\tilde{V}_j(K)$: normalized intrinsic volume)



· Extend the definition of $\tilde{V}_j(K)$ to general closed convex cones by approximation

2nd defn : Let K be convex body, B is unit ball in $\mathbb{R}^d$.

$$\mathrm{vol}(K + tB) = \sum_{j=0}^{d} \binom{d}{j} W_j^{(d)}(K)\, t^j$$

($K + tB$: Minkowski sum; $W_j^{(d)}(K)$: quermassintegrals)

Intrinsic volumes : for $j = 0, 1, \dots d$,

$$V_{d-j}(K) := \frac{1}{\kappa_j} \binom{d}{j} W_j^{(d)}(K)$$

(κ_j: $\mathrm{vol}(B^j)$)

$$\tilde{V}_j(K) := \frac{V_j(K)}{\sum_{j=0}^{d} V_j(K)}$$

Σ_K : intrinsic volume random variable on $\{0, \dots d\}$ with probabilities

- Remark: if K is a closed convex cone, the $tK = K$ $(t > 0) \Rightarrow$ $Z_{tK} \overset{\text{distribution}}{=} Z_K$

But if K is a convex body (compact), since $V_j(tK) = t^j V_j(K)$, we have

$$\lim_{t \to \infty} V_j(tK) = 1\{j = \dim K\}.$$

- Alternative definition of $\delta(K)$:

$$\delta(K) := \mathbb{E}[Z_K]$$

- Thm (Conic kinematic formula) [SW08]
If C, K are c.c.c. in $\mathbb{R}^d$, one of which is not a subspace, and Q random orthogonal matrix, then

$$\mathbb{P}[C \cap Q K \neq \{0\}] = \sum_{i=0}^{d} \sum_{j=0}^{i-1} \left(1 + (-1)^{i-j+1}\right) \widetilde{V}_i(C)\, \widetilde{V}_{d-j}(K)$$

- [ALMT14] seems to contain an incorrect formula

- Concentration of Z_K

"Z_K is close to $\mathbb{E}[Z_K] = \delta(K)$ with high-probability"

If K is a convex body $\Rightarrow Var(Z_K) \leq 4d$

Now,

$$\mathbb{P}[C \cap Q_K \neq \{0\}] \approx \sum_{i=0}^{d} \sum_{j=0}^{i-1} \tilde{V}_i(C)\, \tilde{V}_{d-j}(K)$$

$$= \sum_{i=0}^{d} \sum_{\ell=0}^{d} \tilde{V}_i(C)\, \tilde{V}_\ell(K)\, 1_{i+\ell \geq d+1}$$

$$= \mathbb{E}\left[1_{Z_C + Z_K \geq d+1}\right]$$

($Z_C + Z_K$: independent)

Concentration

$$\approx 1_{\mathbb{E}[Z_C] + \mathbb{E}[Z_K] \geq d+1}$$

($\mathbb{E}[Z_C] + \mathbb{E}[Z_K] = \delta(C) + \delta(K)$)

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